

Accurate Classification of Seizure and Seizure-Free Intervals of Intracranial EEG Signals From Epileptic Patients

Salim Lahmiri¹ and Amir Shmuel

Abstract—Electroencephalogram (EEG) signals are widely used to detect epileptic seizures in a patient's neuronal activity. Since visual inspection and interpretation of EEG signal are time-consuming and prone to errors, various computer-aided diagnostic (CAD) tools have been proposed. In this paper, we present a novel automated detection system to distinguish between intracranial EEG time courses with seizures and those that are seizure-free based on complexity measures. Specifically, the features used to characterize the EEG signals are estimates of multiscaling properties over a large spectrum measured by using the generalized Hurst exponent. We tested the capacity of these estimates to correctly classify seizure intervals using a publicly available data set. Using the k -nearest neighbor classifier and testing with tenfold cross validation, we achieved 100% accurate classification. Our proposed CAD system outperformed the existing state-of-the-art models. Moreover, our CAD system is not only accurate but also fast and simple to implement. Therefore, it can be used as an expert system to support a decision in clinical applications.

Index Terms—Classification, computer-aided diagnosis (CAD), epilepsy, generalized Hurst exponent (GHE), intracranial electroencephalography, k -nearest neighbor (k -NN), long memory, seizure.

I. INTRODUCTION

THE measurement of key salient features in electroencephalogram (EEG) signals is of great importance in biomedical engineering [1]–[4], especially a great help to physicians and scientists for proper characterization of abnormalities in brain electrical activity. Epilepsy is one of these abnormalities. Epilepsy is a neurological condition involving unpredictable seizures. It is a spectrum condition, with a wide range of seizure types that vary from person to person. More than 30% of the patients suffer from seizures that are resistant

to antiepileptic drugs [5]. Uncontrolled epilepsy is harmful to the brain. Patients are at increased risk of sudden death [6] and have a mortality rate of about fivefold higher than that of the general population [7]. Therefore, prompt seizure detection and control of paramount are important.

A growing attention has recently been given to the accurate detection of seizures and epilepsy based on intracranial EEG. EEG is extensively employed for the assessment of epilepsy in clinical applications. For better processing of EEG signals and accurate classification of seizure and seizure-free intervals (SFIs) in epileptic patients, various automatic systems have been proposed based on the analysis of intracranially recorded EEG signals. Extracranial EEG recordings are very sensitive to artifacts and noise and have a poor spatial resolution [8]. Therefore, intracranial EEG signals are more appropriate for the diagnosis. For instance, Joshi *et al.* [9] employed fractional linear prediction (FLP) to model EEG signals. The resulting error energy is then used for signal characterization and training a support vector machine (SVM) classifier with a radial basis function (RBF) kernel. The proposed system achieved 95.33% accuracy. Kaya *et al.* [10] proposed a system that employs a 1-D local binary pattern (1-D-LBP) for feature extraction and BayesNet classifier. The system yielded 97% accuracy, 95% sensitivity, and 98% specificity. Kumar *et al.* [11] used the Gabor filter to process EEG signals, 1-D-LBP for signal segmentation, and features obtained from the histogram of 1-D-LBP to characterize EEG signals. The k -nearest neighbor (k -NN) classifier was used to distinguish between seizure and seizure-free EEG signals. The proposed system achieved the classification accuracy of 98.33%.

In another study, Pachori and Patidar [12] proposed a system that is based on empirical mode decomposition (EMD) to obtain intrinsic mode functions (IMFs). Then, the second-order difference plot (SODP) of each IMF was calculated to provide an elliptical structure. Subsequently, the 95% confidence ellipse area measured from the calculated SODPs was used as a feature to discriminate between seizure and SFIs. The artificial neural network (ANN) classifier achieved 95.75% accuracy. More recently, Tiwari *et al.* [13] employed a pyramid of a difference of the Gaussian filter to process EEG signals to detect key points at multiple scales. Then, LBPs were calculated at each key point. Finally, the feature vector was composed of the histogram of these detected key points.

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The SVM classifier was employed to classify seizure and seizure-free EEG intervals. The authors obtained 99.45 ± 0.25 accuracy based on tenfold cross validation. In addition, Raghu and Sriraam [14] computed four entropy measures from EEG signals, including Shannon, log energy, spectral, and Renyi entropy, which were then used to train a multilayer perceptron neural network for recognition of seizure and SFIs. The obtained accuracy was 84.58%.

Distinguishing between intracranial EEG signals with/without seizures is a difficult problem. Although previous works [9]–[14] achieved high precision in general, ideally, perfect accuracy is needed in clinical settings. In addition, previous works have considerable computational cost in terms of processing time. Therefore, the purpose of our paper is to develop a new automated system to accurately classify seizure and SFIs.

For fast processing and accurate classification of intracranial EEG signals, our proposed automated system uses two stages. In the first stage, the multiscale complexity of EEG signals is measured. Specifically, the Hurst exponent of the EEG signal is computed at different scales so as to obtain the generalized Hurst exponent (GHE) [15] that describes short and long dynamics of the EEG signal. Measuring the multiscale complexity of intracranial EEG signals based on multifractality better fits the problem than using the Tsallis entropy [2], approximate entropy [3], and chaos [4]. In the second step, the k -NN classifier [16], [17] is employed for the classification task.

The goal of the first stage is to characterize EEG signal intervals by capturing their long memory at a different scale of variations. GHE allows a multiscale analysis of the signal under study to determine its scaling properties by computing the q th-order moments of the distribution of the signal's increments, where positive moments describe long variations in dynamics of the underlying signal and negative moments describe short ones.

In addition, the GHE is statistically unbiased and efficient estimator of multiscaling properties [15]. In particular, it provides reliable statistics as it yields much narrower confidence intervals than the multifractal detrended fluctuation analysis, and it does not underestimate the expected value under the presence of heavier tails [18]. Thus, because of these attractive features related to the GHE, the stochastic behavior of the EEG signal can be better described by the level of persistence in its short and long-term variations captured at different scales of analysis. Furthermore, the GHE estimates were found to be effective in the characterization and classification of normal intracranial EEG and extracranial EEG signals with seizure intervals [19], [20]. Moreover, fractal-based estimates were also found to be effective in measuring the development of fetal cortical surface in 3-D magnetic resonance images [22] and classifying Alzheimer's disease, mild cognitive impairment, and normal brain magnetic resonance images [23], [24].

Finally, as indicated previously, we employ the k -NN classifier [16], [17] to distinguish between two categories of intracranial EEG signals: those with seizure and those with SFIs in epileptic patients. The computed q th-order moments by GHE are used to form the features vector for training the

k -NN classifier. It is important to note two important benefits of k -NN. First, it does not assume the form of a fit model [25]; it is entirely based on data-driven learning. Second, in terms of computational complexity, it is efficient and fast. Since GHE estimation and k -NN learning and classification are not over complex, we expect our proposed intracranial EEG signal classification system to also be efficient.

The Gaussian mixture model was previously found to be effective in the classification of neonatal seizures [26] and motor imagery EEG signals [27] thanks to its ability to estimate complex probability density functions from the data set to approximate classes. Specifically, it uses a probabilistic model to distribute the data points into different classes without an *a priori* assumption about the interpretation of the classes. However, here we rely on the k -NN algorithm because of its effectiveness and low computational cost.

The rest of this paper is organized as follows. Section II presents the EEG data set, the GHE, and k -NN classifier. The experimental results are provided in Section III and discussed in Section IV. Section V provides the conclusion.

II. METHODS

We developed an automated system to distinguish between intracranially recorded seizure and SFIs in two stages. In the first stage, GHE is estimated at different scales to characterize the EEG signal by capturing its multiscale long-memory properties. Once computed, GHE estimates are used to train k -NN for classification. In this section, the data set, GHE, and k -NN classifier are presented.

A. Data Set

We used the data set that was also considered in previous studies [9]–[14] to distinguish between EEG signal intervals with and without seizure in epileptic patients. This publicly available data set [28] is provided by the University of Bonn, Germany. It has five subsets: Z, O, N, F, and S, each of which contains 100 single-channel EEG signals. Each signal segment is 23.6 s long and sampled at 173.61 Hz; each EEG signal consists of 4097 samples. All EEG signals were recorded by using a 128-channel amplifier-based system with a common average reference. As in [9]–[14], here we focus only on subsets N, F, and S, which were recorded intracranially. In particular, N and F are seizure-free EEG signals recorded in the epileptogenic zone and hippocampal formation, respectively. Subset S contains EEG signals recorded during the seizure activity. Fig. 1 displays 200-sample EEG signals from sets N, F, and S for illustration purpose.

We first grouped subsets N and F to form one class of SFIs, as was also done in previous studies [9]–[14]. Note that the original intervals were not manually processed; simply a new set of data made of the SFI subsets N and F is compared to the subset S.

B. Generalized Hurst Exponent

The GHE [15] is employed to measure the complexity of EEG signals by estimating their multiscaling (multifractal)

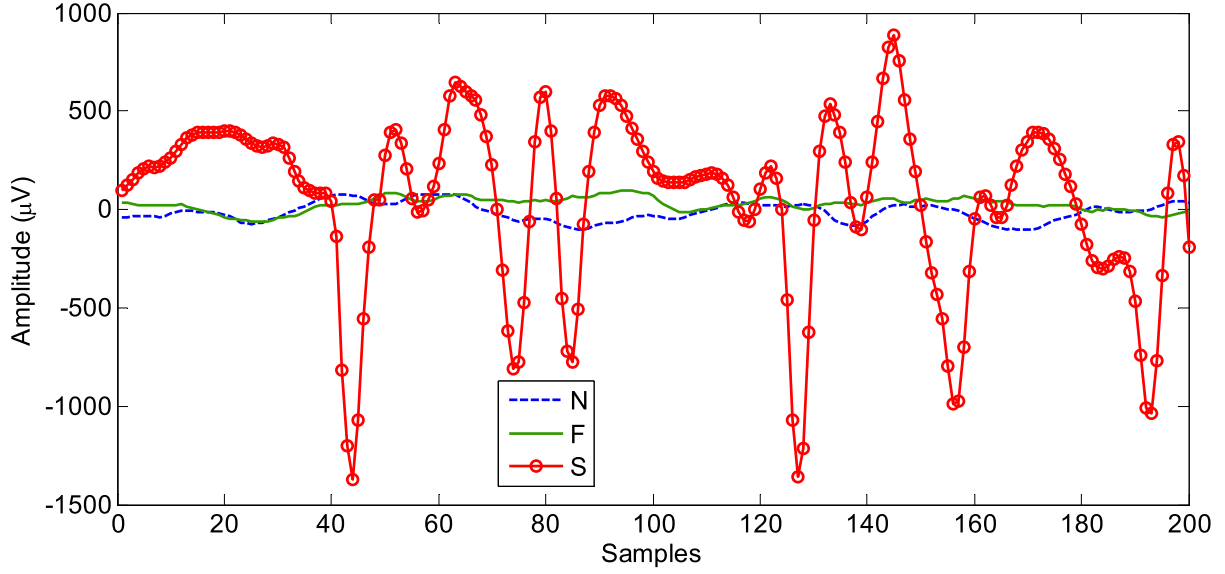


Fig. 1. Three examples of EEG signals from subsets N (blue dashed line), F (green solid line), and S (red circles). Both N and F contain epileptic SFIs. Subset S contains seizure intervals. The original EEG segment is 23.6 s long and sampled at 173.61 Hz yielding to 4097 samples (measurements). Here, we present the first 200 samples.

properties. These measurements allow an appropriate characterization of nonlinear dynamics of EEG signals. In particular, long memory of the EEG is computed at different scales to describe its short and long variations. Consider a signal $S(t)$ defined at discrete-time intervals $t = v, 2v, \dots, T$ over a period T that is an integer multiple of v . Then, the q th-order moment of the distribution that describes the statistical evolution of $S(t)$ is given as [15]

$$K_q(d) = \frac{\langle \|S(t+d) - S(t)\|^q \rangle}{\langle |S(t)|^q \rangle} \quad (1)$$

where $d \in [v, d_{\max}]$ is the time interval and $d_{\max}$ is its predetermined upper limit. The GHE given by $H(q)$ is defined by the scaling behavior of $K_q(d)$ based on the following empirical relation:

$$K_q(d) \propto \left(\frac{d}{v}\right)^{qH(q)}. \quad (2)$$

If $K_q(d)$ and d satisfy a linear relationship for a given order q in the log-log scale, the Hurst exponent $H(q)$ can be estimated by applying a linear regression of $\log(K_q(d))$ versus $\log(d)$. Thus, $H(q)$ describes long-memory dependence or persistence in the original signal $S(t)$.

Specifically, the multiscaling structure of signal $S(t)$ is related to different orders q of the exponent $H(q)$. As a result, $H(q) > 0.5$ indicates that the signal fluctuations associated with order q are persistent. On the contrary, $H(q) < 0.5$ indicates that the signal fluctuations associated with order q are antipersistent. Finally, the signal fluctuations follow a random walk as $H(q) = 0.5$. In our study, the range of q is fixed to the interval from -10 to 10 ($q \neq 0$) to cover a large width. The parameters v and $d_{\max}$ are, respectively, set to 5 and 19. A feature vector is constructed. It includes all estimated $H(q)$, where $-10 \leq q \leq 10$ and $q \neq 0$. The obtained feature vector is fed to the k -NN classifier for training

and classifying intracranial EEG signals of epileptic patients: those with seizure against those with SFIs.

C. k -NN Classifier

The k -NN algorithm [16], [17] is a nonparametric supervised classifier in which retrieving nearest neighbors is based on similarity. It clusters a set of data points into groups and classifies new data based on a measure of similarity (such as the Euclidean distance). Given a value k and a feature vector to classify I , it locates the k -NNs of I in the sample set and uses the categories of neighbors to determine the class of I . This structure imposes a low computational burden. Therefore, the k -NN algorithm is fast. The common algorithm of k -NN is as follows.

- 1) Compute the Euclidean distances between a new object o and all the objects in the set used for learning.
- 2) Choose the k objects from the learning set, which are closest to o .
- 3) Classify o to the group associated with the largest number of objects from the k objects.

After locating the k -NNs, the class of the new object is identified by using a voting algorithm such as majority voting or weighted-sum voting [17]. In this paper, we implemented the simple majority voting algorithm. In summary, the formal k -NN classifier algorithm is as follows [25]:

$$\arg \min(d_e(t, o, k)) \Rightarrow \text{identify } P \quad (3)$$

where t is the training data, o is the object to be classified, P is the assigned class of the new object, k is the number of closest neighbors to be considered, and d_e is the Euclidean distance given as

$$d_e(t, o, k) = \sqrt{\sum_{i=1}^L (t_{i,k} - o_{i,k})^2} \quad (4)$$

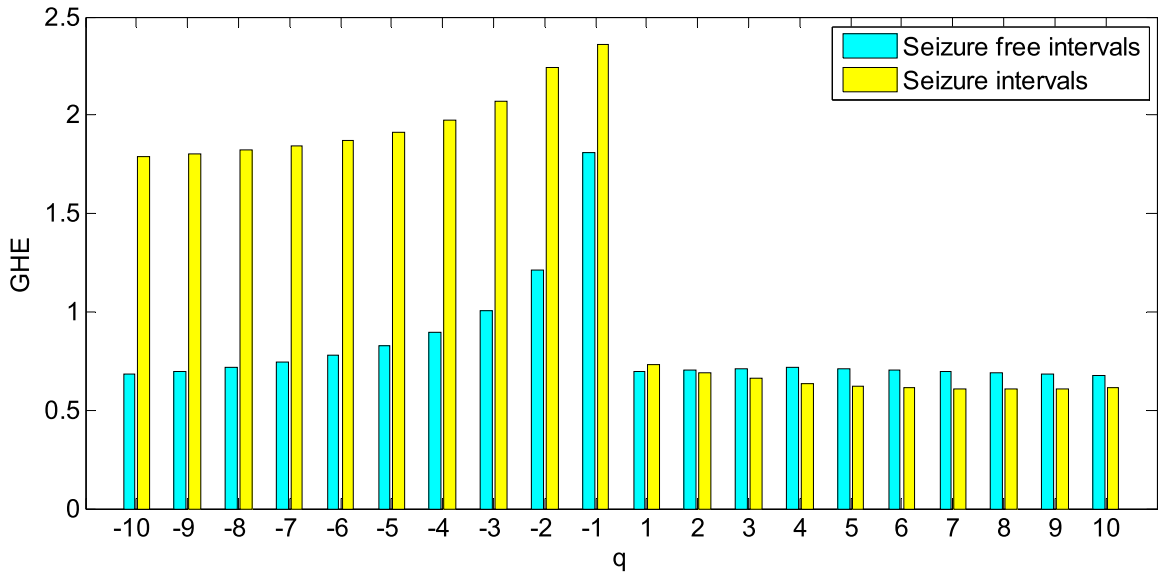


Fig. 2. Multiscale behavior of GHE as a function of scale q .

TABLE I
RESULTS FROM STATISTICAL TESTS

Statistical tests	t -test/ F -statistic/ χ^2	p -values
t -test	-3.7225	6.3715×10^{-04}
F -test	0.1905	6.9553×10^{-04}
Kruskal-Wallis test	5.0407	0.0248
The statistics associated with t -test (t -statistic), F -test (F -statistic), and Kruskal-Wallis test (χ^2). All statistical tests were performed at the 5% significance level.		

where L is the length of each data vector. In this paper, the parameter k is set to one, since this is a conservative value that allows considering only the closest pattern.

III. EXPERIMENTAL RESULTS

In order to assess the performance of k -NN trained with estimated values of GHE [$H(q)$], a tenfold cross validation is adopted and performed 20 times to assure more repetitions and randomness. The resulting average values of accuracy, sensitivity, and specificity are used as performance metrics. Fig. 2 shows the average multiscale behavior of GHE as a function of scale q for all 200 EEG signals with SFIs (subsets N and F) and 100 EEG signals with seizure intervals (subset S). For negative values of q , the values of GHE of EEG signals with seizure intervals are much higher than those of EEG signals with SFIs. In contrast, for positive values of q (except for $q = 1$), the values of GHE of EEG signals with seizure intervals are lower than those of EEG signals with SFIs. These preliminary conclusions are formally confirmed after having performed a two-sample t -test, two-sample F -test, and two-sample Kruskal-Wallis test, all at 5% statistical significance level. The first, second, and third tests were applied in order to check whether SFIs and epileptic seizure intervals (ESIs) have similar means, variances, and distributions, respectively. Table I presents the results: the means, variances, and distributions of GHEs associated with SFI and ESI are statistically different. Therefore, they are

TABLE II
CORRECT CLASSIFICATION RATES OBTAINED BY OTHER TECHNIQUES USING THE SAME DATA SET

Study	Approach	Accuracy
[9]	FLP+RBF SVM	95.33%
[10]	1D-LBP+BayesNet	97%
[11]	Gabor filter+ 1D-LBP+Histogram+ k -NN	98.33%
[12]	EMD+SODP+ANN	95.75%
[13]	Gaussian filter+LBP+Histogram+SVM	99.45%±0.25
[14]	Entropies+ MLP-NN	84.58%
Our work	GHE+ k -NN	100%

distinct features making it possible to distinguish between SFIs and ESIs.

Indeed, the tenfold cross validation showed that the proposed classification system achieved 100% accuracy, indicating that the set of features composed of 20 estimated GHEs at different scales can differentiate seizure intervals and SFIs. The sensitivity and specificity performance measures were also 100%. Table II reports the classification rates obtained by the state-of-the-art methods [9]–[14] using the same data set on exactly the same problem: distinguishing between seizure and SFIs in intracranial EEG signals of epileptic patients. Our proposed system outperformed all the existing state-of-the-art models [9]–[14].

IV. DISCUSSION

We have presented an automated seizure detection system in intracranial EEG signals based on a novel extracted feature used to characterize the original EEG signal multiscale properties and the k -NN classifier. We rely on estimating complexity by means of fractals in intracranial EEG records as it is an effective descriptor of a given signal.

Although there are various techniques that can be used to measure complexity in EEG signals such as the Tsallis entropy [2], approximate entropy [3], and chaos [4], we selected GHE because of its capacity to measure

complexity at different scales. This is achieved by estimating multifractality in the EEG signal such that long-range correlations in its short and long variations are appropriately measured. In this regard, they were found to be statistically and significantly different, differentiating SFIs and those with seizure intervals based on three statistical tests, namely, t -test, F -test, and Kruskal–Wallis test.

There are two main differences between this paper and that in [20]. First, in [20], GHE-based estimates were used to distinguish between normal extracranial EEG records and intracranial EEG records with seizure intervals. In contrast, this paper focuses on distinguishing between seizure and SFIs recorded exclusively from intracranial electrodes implanted in epileptic patients. Other studies using a tunable- Q factor wavelet transform and bootstrap aggregating [29], complete ensemble EMD with adaptive noise (CEEMDAN)-based features and programming boosting [30], and CEEMDAN-based features and neural networks [31] have not examined the specific problem of classifying seizure and SFIs in intracranial EEG signals of epileptic patients. For clear ethical and safety reasons, obtaining intracranial EEG records from healthy subjects is impossible. Therefore, it is impossible to compare intracranial EEG obtained from patients and healthy subjects. Second, our study sheds light on the nonlinear dynamics of seizure intervals. Indeed, it is crucial to understand the nonlinear dynamics and statistical properties of intracranial EEG signals in epileptic patients for better prediction and treatment of the seizure.

Furthermore, the accuracy achieved by our proposed system is higher than the best accuracy reported by the state-of-the-art methods [9]–[14] applied to the same data set on the same problem: classification of seizure and SFIs recorded from intracranial electrodes implanted in epileptic patients. This shows that GHE estimates create a significant discriminative space for an accurate classification of seizure intervals and SFIs. This means that the multiscale self-similarity of intracranial EEG signals is a powerful feature to characterize them; the complexity of EEG signals is altered by seizures.

In addition, our automated system for the EEG signal analysis and classification does not lead to overhead computing as GHE estimation and k -NN learning and testing are efficient. Indeed, the feature extraction and classification task stages took altogether less than 7 s to perform on a PC station with a 2.1-GHz Core Duo CPU in the Matlab environment. The proposed intracranial EEG signal classification is straightforward to implement, fast to execute, simple to interpret, and accurate. Therefore, it can be employed as an expert system for detecting intracranial seizure intervals.

V. CONCLUSION

We propose a new automatic system for intracranial EEG signal classification as a mechanism to accurately distinguish between seizure intervals and SFIs in epileptic patients. It is based on multiscale properties captured by GHE estimates. Our analysis shows that our newly proposed system outperforms the best existing strategies in terms of classification accuracy. In the future work, we will extend our approach to the multiclassification problem of EEG signals and

investigate whether GHE-based estimates can be employed in an automated seizure prediction system.

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